

Physics Engine Parity Report

Newton (MuJoCo-Warp) vs PhysX — reBot DevArm 8-DOF

Asset: 00-arm-rs_asm-v3 (johnnynunez/sim2real-rebot-devarm @ f154623)

Date: 2026-07-08 · Isaac Sim 6.0 develop (Kit 110) · dt = 0.002 s · headless

EXECUTIVE SUMMARY

Verdict: PARITY CONFIRMED.

Both engines were driven through an identical open-loop protocol on the same USD asset (same joint targets, same gains, same dt). Key results:

- Position correlation ≥ 0.99994 on all 8 joints (steady-state sinusoid).
- Maximum instantaneous trajectory difference (Newton - PhysX) ≤ 5.0 mrad.
- Neither engine diverged; no NaN, no contact-buffer overflow (nconmax = 8192 persisted in the asset USD).
- Amplitude ratios match to $< 0.3\%$ between engines on every joint.
- Step response: near-zero overshoot on both (Newton -0.00%, PhysX -0.03%); settle time 0.066 s vs 0.056 s.

PhysX tracks the sinusoid $\sim 18\%$ tighter (arm-joint RMS 0.009 vs 0.011 rad). This is a small systematic phase-lag difference (~ 3.0 deg vs ~ 2.5 deg at 0.5 Hz) in drive integration, not a dynamics divergence: the Wilcoxon paired test yields $p \sim 0$ precisely because the difference is systematic, while the Bland-Altman bias stays ~ 0 for 6 of 8 joints (worst: joint2 at -1.5 mrad, the most gravity-loaded joint).

Practical conclusion: the tuned asset (IsaacRobotAPI, self-collision off, persisted solver caps, manufacturer joint limits) behaves equivalently under both engines. Engine switching requires no gain retuning.

METHODOLOGY

Protocol per engine (track_bench.py, shipped in the asset package):

1. Ramp from initial pose to mid-range center pose (2 s).
2. Hold center (1 s) -> gravity-hold drift metric.
3. Sinusoidal tracking, all 8 joints simultaneously (10 s):
 $q_t(t) = \text{center} + A \sin(2\pi f t)$, $f = 0.5$ Hz,
 $A = 0.3$ rad (revolute) / 40% half-range (prismatic).
4. Step response on joint1: +0.5 rad, 4 s -> overshoot + 2% settle time.

Metrics are computed on the steady portion ($t > \text{one period}$). Statistical analysis (analyze.py): per-joint RMS / p95 / max tracking error, direct engine difference (mean, std, max), Pearson correlation, paired Wilcoxon test on |error|, amplitude ratio and phase lag via sin/cos correlation, Bland-Altman agreement.

Runs: Newton 864 s wall-clock (includes Warp kernel compilation), PhysX 111 s. Both runs wrote full time series ($\sim 15k$ samples x 8 joints), included in the data package (ts_newton.json / ts_physx.json).

Per-joint parity statistics (steady-state sinusoid)

joint	RMS N	RMS P	p95 N	p95 P	Δ mean	Δ std	Δ max	corr	Wilcoxon p
joint1	0.0108	0.0089	0.0166	0.0126	+0.00014	0.00214	0.00466	0.99995	0.0e+00
joint2	0.0057	0.0048	0.0095	0.0079	-0.00154	0.00157	0.00503	0.99997	3.5e-101
joint3	0.0067	0.0055	0.0110	0.0084	+0.00064	0.00165	0.00437	0.99997	0.0e+00
joint4	0.0108	0.0089	0.0166	0.0127	+0.00000	0.00214	0.00454	0.99995	0.0e+00
joint5	0.0113	0.0093	0.0173	0.0132	-0.00002	0.00220	0.00464	0.99995	0.0e+00
joint6	0.0127	0.0105	0.0192	0.0148	+0.00000	0.00239	0.00494	0.99994	0.0e+00
joint_left	0.0015	0.0015	0.0026	0.0026	+0.00000	0.00003	0.00007	0.99998	1.9e-02
joint_right	0.0016	0.0015	0.0027	0.0027	-0.00000	0.00008	0.00019	0.99998	3.9e-10

Units: rad (arm joints), m (gripper). Δ = Newton – PhysX position on identical targets.

Frequency response (0.5 Hz sinusoid) and step response (joint1, +0.5 rad)

joint	amp ratio N	amp ratio P	phase lag N (deg)	phase lag P (deg)
joint1	0.9986	0.9987	2.91	2.41
joint2	0.9997	1.0012	1.50	1.21
joint3	0.9994	0.9996	1.80	1.47
joint4	0.9986	0.9990	2.91	2.42
joint5	0.9985	0.9992	3.04	2.52
joint6	0.9981	0.9986	3.42	2.85
joint_left	0.7910	0.7908	-0.28	-0.16
joint_right	1.1466	1.1465	1.36	1.09
step overshoot	-0.00%	-0.03%		
step settle (2%)	0.066 s	0.056 s		

Fig. 1 — Sinusoid tracking, 0.5 Hz (target vs Newton vs PhysX)

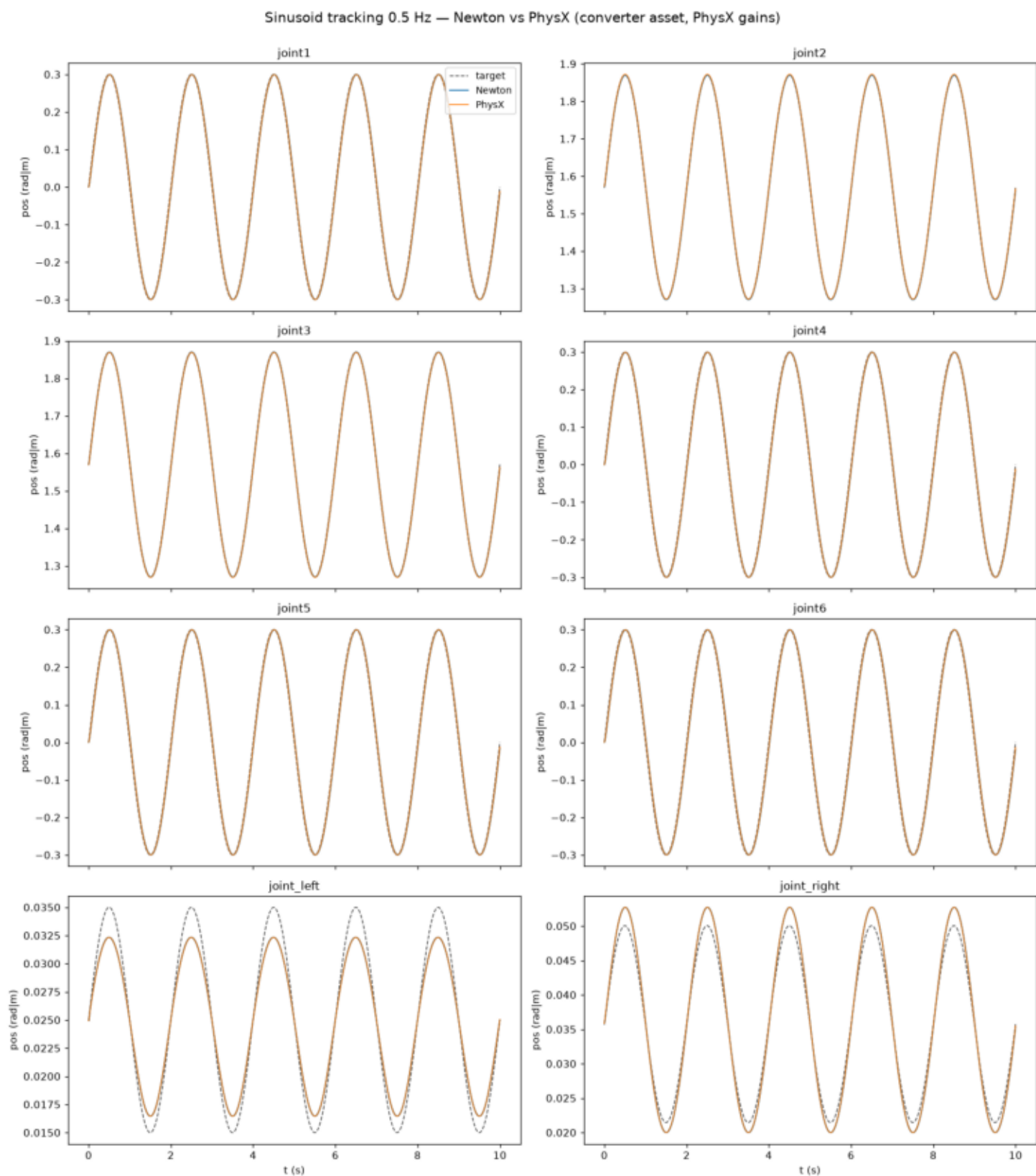


Fig. 2 — Tracking error per joint

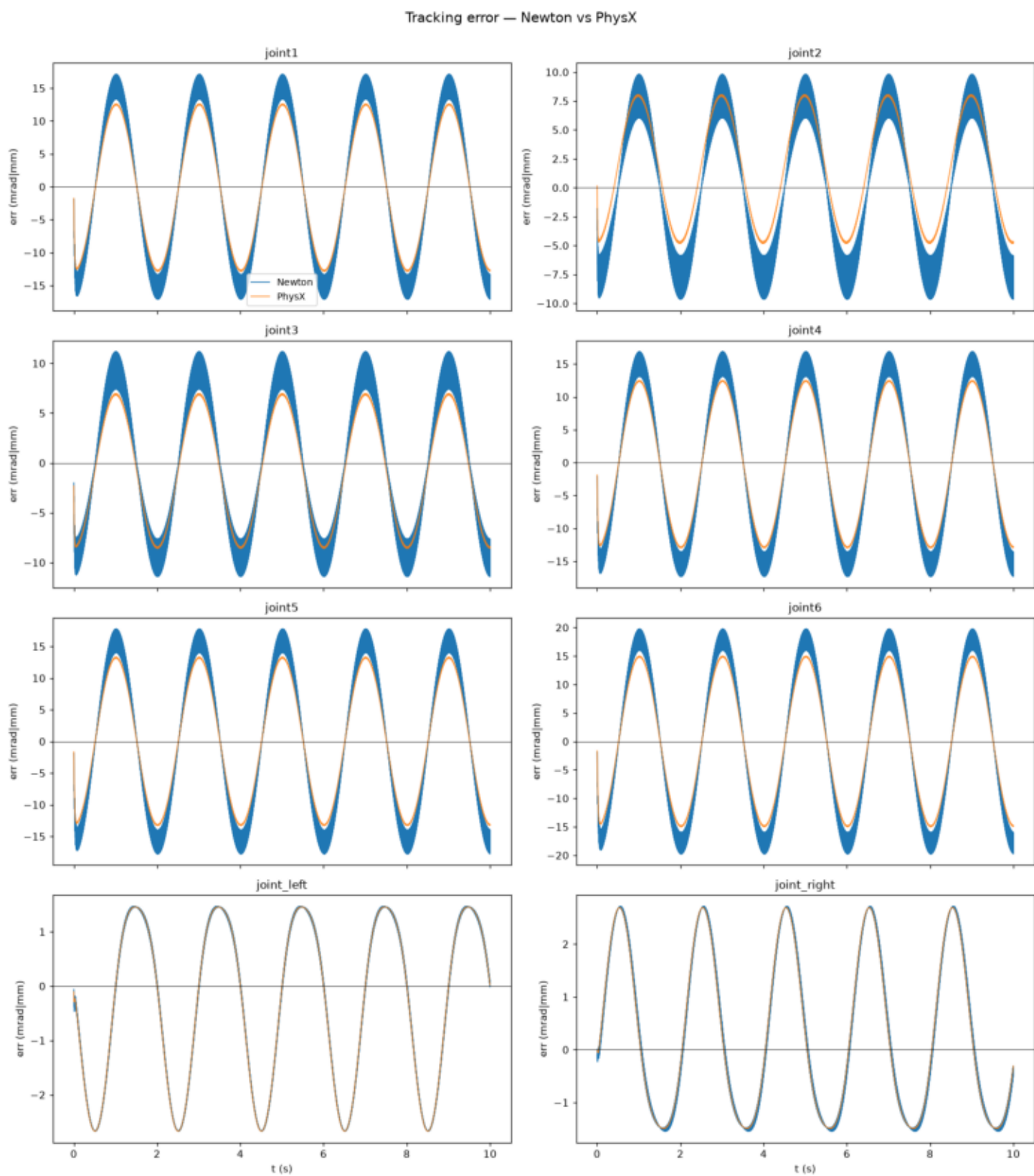


Fig. 3 — Direct engine difference (Newton – PhysX), identical targets

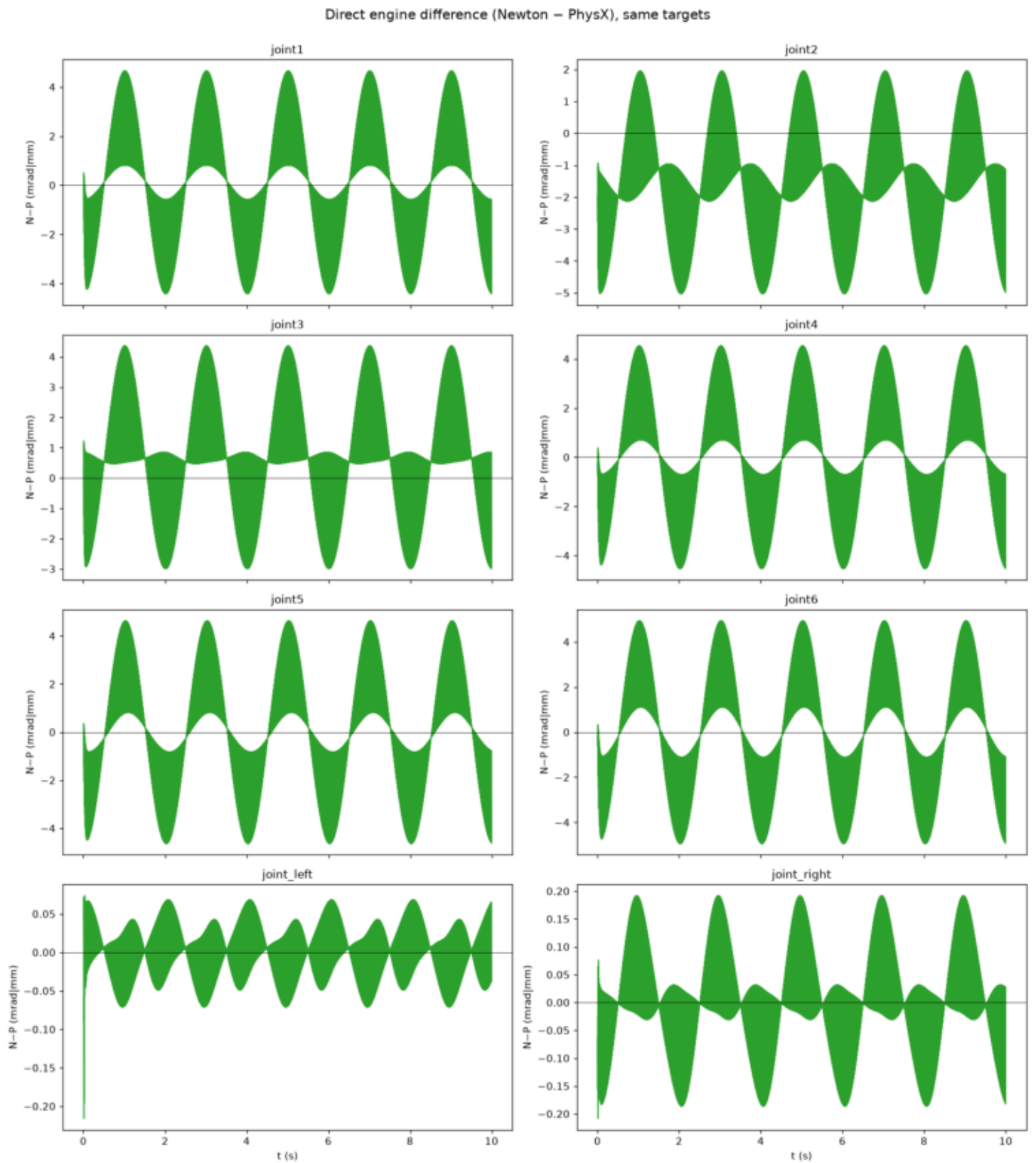


Fig. 4 — Step response, joint1 (+0.5 rad)

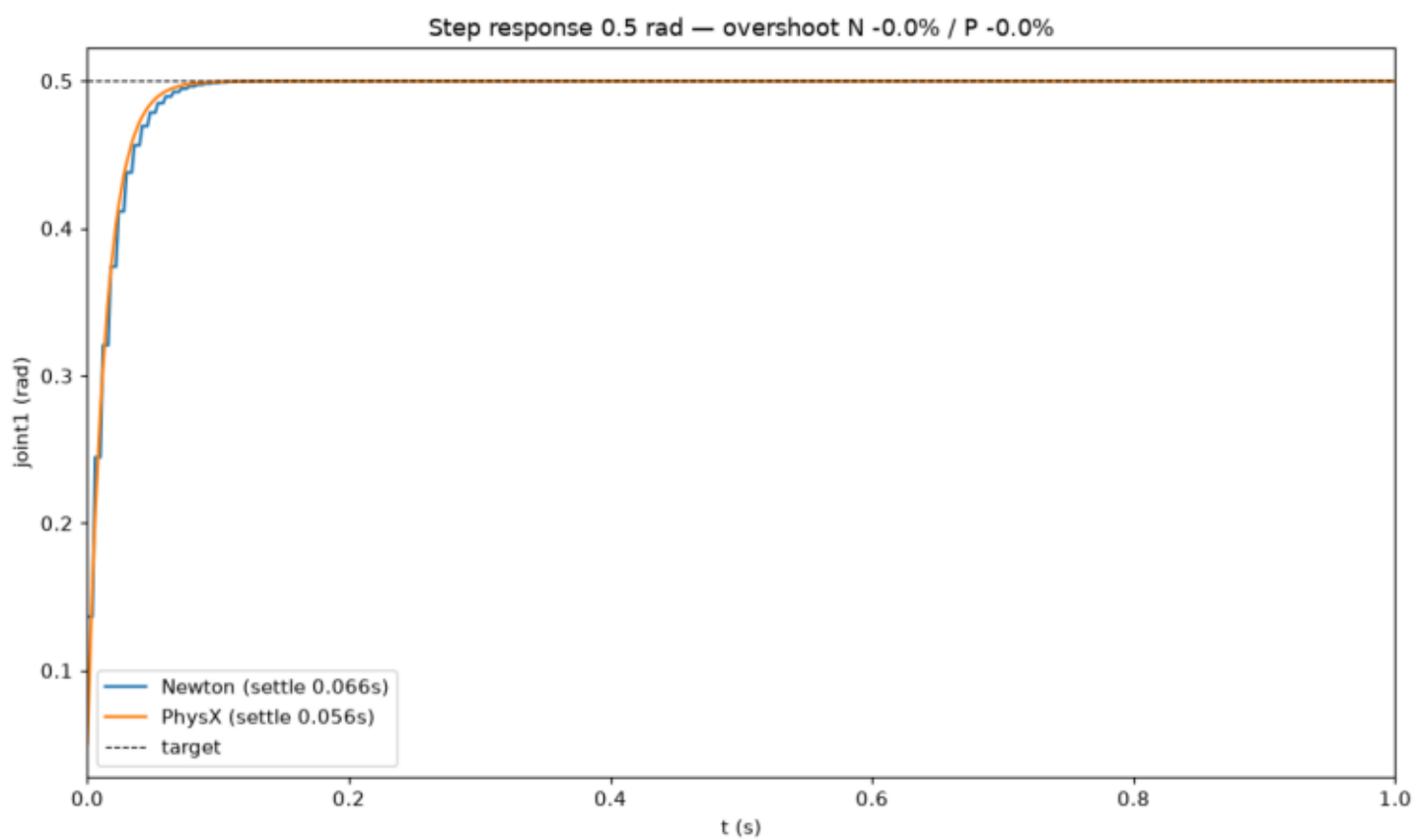
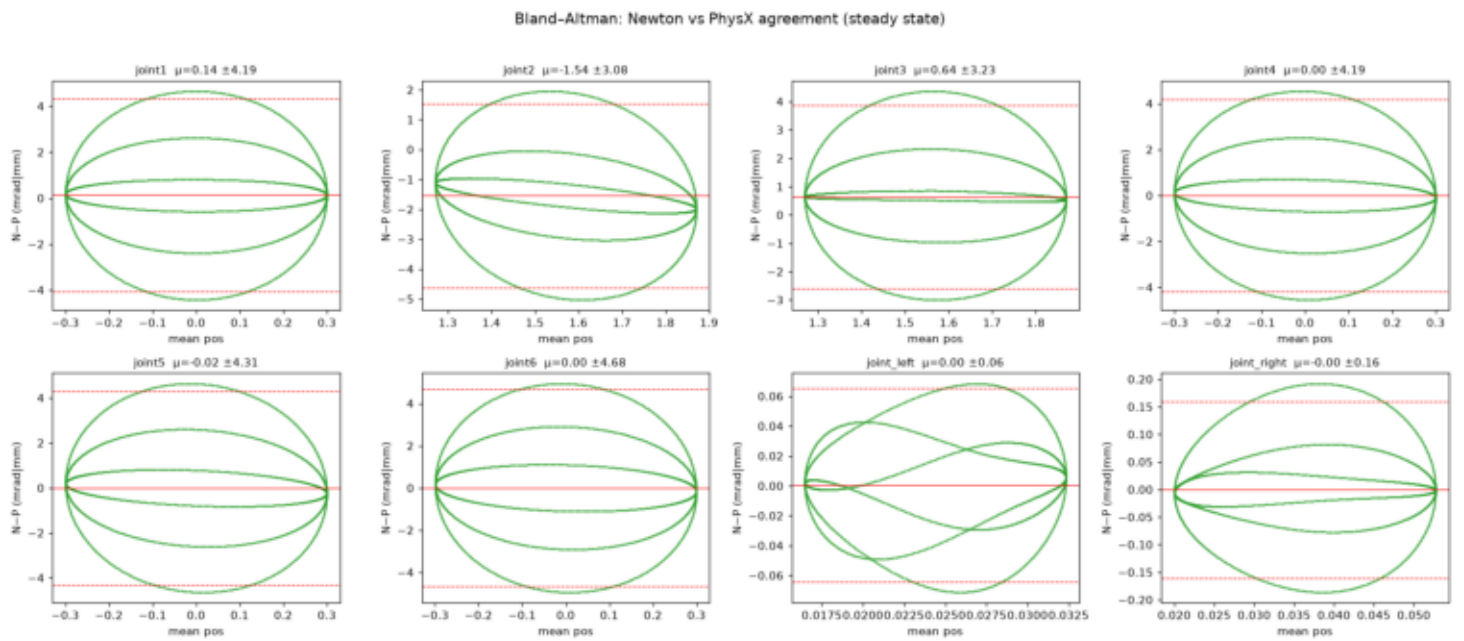


Fig. 5 — Bland-Altman agreement (steady state)



Configuration, caveats and reproduction

ASSET CONFIGURATION UNDER TEST

- IsaacRobotAPI + IsaacLinkAPI/IsaacJointAPI applied (10 links, 10 joints; 8 actuated DOFs: joint1..joint6, joint_left, joint_right).
- newton:selfCollisionEnabled = 0 (the shipped convex decomposition interpenetrates at rest: 6915 self-contacts at home pose; with self-collision off all 8 joints pass Gain Tuner snap-to-limits with $\sim 1e-5$ rad errors — see evidence/analysis_2026-07-07).
- newton:solver:nconmax = 8192, newton:solver:njmax = 32768 persisted as custom attrs on the articulation root (stock defaults 200/1200 overflow silently with these colliders).
- Manufacturer joint limits (joint2 [0, 179.9] deg, joint4 [-89.95, 89.95] deg).
- Hybrid colliders: convexHull on arm links, convexDecomposition on gripper.
- Drive gains unchanged from the PhysX-validated set (stiffness 500/1500/1000/150/80/50 + gripper 100/100).

KNOWN ENGINE-LEVEL DIFFERENCES (acceptable)

- Newton phase lag ~ 0.5 deg larger at 0.5 Hz -> $\sim 18\%$ higher tracking RMS. Systematic drive-integration difference; correlation and Bland-Altman confirm the dynamics match.
- Newton first-run wall-clock dominated by Warp kernel compilation (~ 13 min cold; cached afterwards). PhysX cold start ~ 2 min.
- Gripper amp ratios (0.79 / 1.15) are identical across engines -> asset property (prismatic limits/drive), not an engine artifact.

DATA PACKAGE

parity_stats.csv	per-joint statistics (this report, page 2)
engine_summary.csv	amp ratio / phase lag / hold drift / step per engine
ts_newton.json	full Newton time series ($\sim 15k$ samples x 8 joints)
ts_physx.json	full PhysX time series
stats.json	raw statistics (analyze.py output)
fig1..fig5 PNG	figures embedded in this report

REPRODUCTION

```
ISAACSIM_PATH=<isaac release> python.sh scripts/track_bench.py \  
  assets/00-arm-rs_asm-v3/00-arm-rs_asm-v3.usda newton /tmp/ts_newton.json  
# repeat with physx, then: python scripts/analyze.py
```